DATAX500

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Assignment 1

1. Using the '*heart.csv*' data-set, calculate the following statistical measures for age in *R* **(6 marks)**

- First, we Load the data-set:

```
# load automatically via git files
heart<-read.csv("Heart.csv")


## Alternatively, load the dataset manually using:
# heart<-read.csv(file.choose())
```

a) Mean **(2 marks)**

```
# assign age column to "age" var for easier processing
age<-heart$age

# calculate the mean age
mean_age<-mean(age)
mean_age
```

```
## [1] 54.01038
```

```
# round to 2 d.p.
mean_age<-round(mean_age, 2)
mean_age
```

```
## [1] 54.01
```

b) Median **(2 marks)**

```
# calculate the median age
median_age<-median(age)
median_age
```

```
## [1] 54
```

c) Sample standard deviation **(2 marks)**

```r
# calculate the sample standard deviation
sd(age)
```

```
## [1] 9.132316
```

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2. A researcher wishes to investigate the relationship between gender and cholesterol levels.

   Using the dataset, perform an appropriate statistical test in R and comment on the output. Your answer should include a null and alternative hypothesis, as well as a p-value **(5 marks)**

   - Since gender (categorical) and cholesterol (continuous) are independent from eachother, we conduct an independent (two-sampled) t-test. But first, we define some hypotheses:

     Null hypothesis: -
   - Alternative hypothesis:
   - p-value:

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3. A researcher wishes to investigate the predictors of maximum heart rate. Using the dataset provided, fit an apprpopriate statistical model of your choice. Make sure to include age as one of the predictors **(14 marks)**

   a) What are the null and alternative (two-sided) hypotheses associated with age? **(2 marks)**

   b) What is the estimated coefficient of age and what does this imply in terms of the relationship between age and maximum heart rate? **(3 marks)**

   c) Comment on the predictors you have included in the model and justify the inclusion of these variables in your model **(3 marks)**

   d) interpret the model output in plain English. Your interpretation must include the p-value associated with each coefficient (slopes and intercept) test, and R-squared value **(3 marks)**

   e) Comment on whether the assumptions of a linear regression model have been met. Include appropriate statistical plots to support your answer **(3 marks)**

Data dictionary:

cp: chest pain type

1. typical angina
2. atypical angina
3. non-anginal pain
4. asymptomatic

trtbps:

- resting blood pressure (in mm Hg)

chol:

- cholesterol (in mg/dl)

gender:

1. female;
2. male;

fbs: (fasting blood sugar > 120 mg/dl)

0. False
1. True

rest_ecg: resting electrocardiographic result

0. Normal
1. Having ST-T wave abnormality (T wave inversions and/or ST elevation or depression of > 0.05 mV)
2. Showing probably or definite left ventricular hypertrophy by Estes' criteria

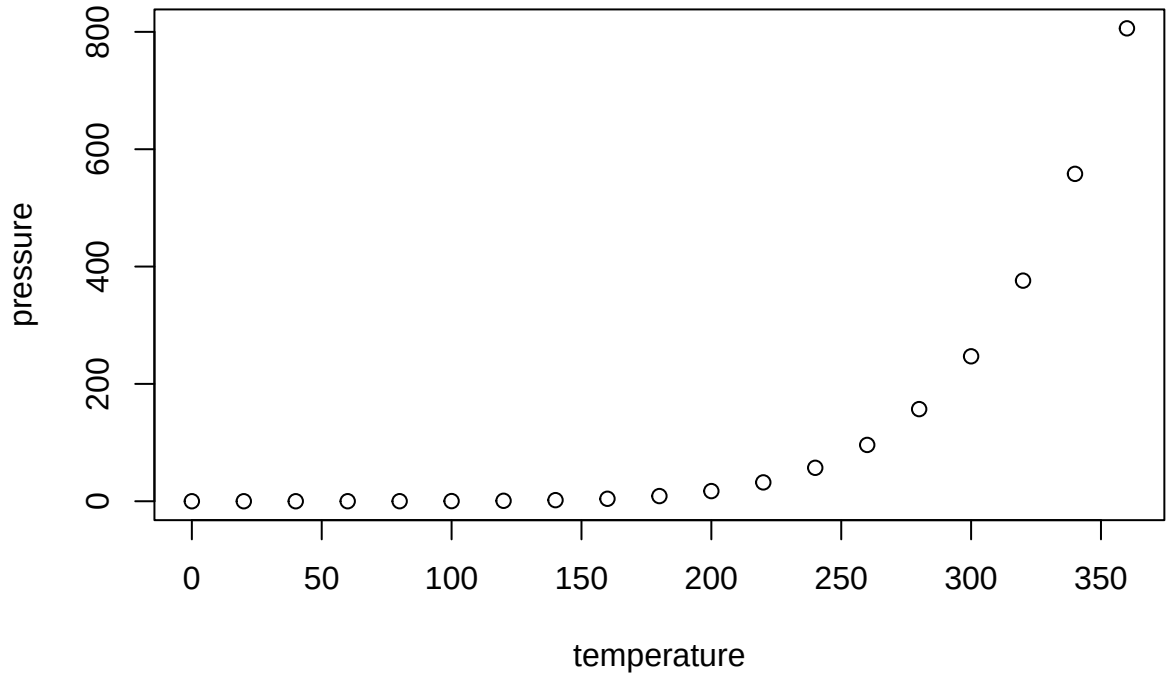
---

```r
summary(cars)
```

```
##      speed           dist       
##  Min.   : 4.0   Min.   :  2.00  
##  1st Qu.:12.0   1st Qu.: 26.00  
##  Median :15.0   Median : 36.00  
##  Mean   :15.4   Mean   : 42.98  
##  3rd Qu.:19.0   3rd Qu.: 56.00  
##  Max.   :25.0   Max.   :120.00  
```

Including Plots

You can also embed plots, for example:



Note that the `echo = FALSE` parameter was added to the code chunk to prevent printing of the R code that generated the plot.